Natural-Language Controls for Short-Video Recommenders

An offline study of controllability

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Experiment Two results

100 fresh synthetic requests.

Commands compared with model-generated references.

Llama: 76 exact, 23 rejected, 1 accepted mismatch.

Rules: 11/100 exact matches.

Requested counts plus ranked exposure change:

20.145% Llama vs 3.815% item feedback.

Conditional requests: no control successes.

NDCG@10: 0.02580 vs 0.03184.

Research question

Can one natural-language interaction satisfy a richer intended change than one item reaction?

- A compound request can specify several categories, strength and exceptions.
- The LLM must also outperform a rule parser to establish its own contribution.
- Offline compliance and historical prediction measure different outcomes.

The claim that language *feels more natural* requires human evidence beyond this study.

Prior work supports a control question

Mozilla (2022): Dislike prevented 12% of unwanted recommendations; Not interested prevented 11%.

Ramos et al. (2024): positive profile edits raised Coverage@10; RMSE 0.941 vs MF 0.925 (lower better).

Sanner et al. (2023): language preferences were competitive near movie cold start.

Chen and Pu (2012): richer feedback through critiquing.

These findings come from different settings.

Full source details: paper/references.bib.

Translation and enforcement are separate

Stage	Role
Recommender	Frozen SASRec-style scores for the candidate catalogue.
Translator	Llama 3.1 8B or rules compile language into a validated command.
Enforcer	Shared deterministic reranking applies the command.
Explanation	Verified facts describe the displayed item and control.

A correct hard exclusion comes from the enforcement code. It does not demonstrate language understanding.

Data and information boundaries

- KuaiRec cohort: **865 users**, **2,593 candidates**, 31 opaque category identifiers.
- Positive event: watch ratio > 2, an engagement proxy.
- Global chronological windows: nominal 70% training, 15% request history, 15% held-out evaluation.
- Requests use training and request history only.
- Quality target: first eligible held-out positive item.

Synthetic requests and numeric category labels limit claims about real preferences and language.

Filtered cohort counts, not the upstream KuaiRec matrix size. Frozen candidate policy permits seen items.

Five matched configurations

ID	Mechanism
A	One simulated reaction changes one historical exemplar item.
B	A plus display-only explanations.
C	Free text compiles into a category control.
D	An editable preference sentence compiles into a category control.
E	Text, profile and explanations, with duplicate commands removed.

One boost and one mute request per user, under seeds 42, 43 and 44. The item-feedback proxy is an explicit design choice.

A is not a reconstruction of a commercial feedback algorithm. B must rank identically to A.

Parser results on synthetic language

Language material	Llama correct	Meaning
Templated inputs	120/120	Canonical task
Broader synthetic set	79/100	Below 85% target
Adversarial subset	2/10	Major weakness

- 79% exceeds the 70% floor. It misses the 85% target.
- The rule parser scores 81/100 on the same broader set.
- The 120 canonical wording patterns informed development.

The adversarial subset belongs to the 100 cases. These finite synthetic sets do not establish general language reliability.

Category enforcement works under the policy

Top-10 exposure	A	C, D and E
Boost target share, TCP	11.6%	79.2%
Muted target share, TCER	6.1%	0.0%

- Hard exclusion guarantees zero exposure when parsed correctly.
- Category-wide boosts affect more items than A's single-item adjustment.
- Rule and LLM paths produce identical feeds for this task.

Averages over 865 users and three seeds. This is enforcement evidence, not an LLM advantage.

Steering has a predictive quality cost

Boost requests	A	C
NDCG@10	0.03012	0.01637

45.7% relative decrease.

Paired mean change: -0.01375

95% user-bootstrap interval: $[-0.02086, -0.00655]$.

This post-hoc secondary contrast measures agreement with a historical next item. It does not measure user dissatisfaction after steering.

Post-hoc exploratory interval, outside the frozen primary BH family.

A weak backbone limits the quality claim

Uncontrolled ranking	HR@1
SASRec, seed 42	0.000%
SASRec, seed 43	0.347%
SASRec, seed 44	0.462%
Recency popularity	7.977%

Recency pools all users' request history. The model's weights use the older training window. This comparison is descriptive. The absolute 10 percentage point non-inferiority margin is uninformative at this baseline.

Post-hoc uncontrolled baselines. A pooled HR@1 interval includes zero. No accuracy gain is established.

Experiment 1 conclusion

1. Deterministic category control works: TCP 11.6% to 79.2%, TCER 6.1% to zero.
2. Llama parsed 120/120 templated inputs, 79/100 broader synthetic inputs and 2/10 adversarial cases.
3. All 10/10 explanation fact selections equal the template.
4. Boost NDCG@10 falls 45.7%. Non-inferiority is uninformative.
5. Backbone HR@1 is 0.0–0.46%, below recency popularity's 8.0%.

The study establishes bounded enforcement and its limitations. It does not establish an LLM advantage or useful quality preservation.

What the fresh experiment tests

Control success versus rules and item feedback.

Requested exposure movement versus item feedback.

Historical next-item accuracy as a separate outcome.

The 73/100 practice bank was already exposed.

Synthetic requests do not measure human experience.

Four classes of intent

Class	Illustrative request	Top-10 requirement
Compound	More X and less Y	$n(X) \geq 2$, $n(Y) \leq 1$
Graded	A bit more X	$2 \leq n(X) \leq 4$
	Much less X / none of X	$n(X) \leq 1$ / $n(X) = 0$
Floor	Less X, but not zero	$1 \leq n(X) \leq 2$
Conditional	More X only if also Y	$n(X \cap Y) \geq 2$ and no increase in $X \setminus Y$ count

Success also requires a full list and positive intended movement. Bands are operational conventions.

X and Y denote opaque categories. Conditional means the same item carries both labels. These are illustrative wording forms, not reserved cases.

A matched interaction budget

- A gets one positive or negative reaction on one historical exemplar.
- Rule and LLM get the complete utterance once and share the same ranker.
- Assignments use only earlier history and a committed SHA-256 scheme.
- Channels share permitted pre-test information. Recency and SASRec use that information differently.

The budget matches interaction count. It does not equalize semantic content, user effort or every possible feedback algorithm.

Prospective design

Protocol and source frozen before fresh requests.

100 balanced requests across four classes.

Two separate model annotation passes.

Original labels locked before comparison.

Agreement measures model consistency, not human validity.

Both passes used the same reported model family.

Fresh bank and references

100 requests; 25 in each of four classes.

31 target category identifiers.

Screened against 100 exposed and 48 development texts.

Model-label agreement: 100/100; kappa 1.0000.

No adjudication or selected close paraphrase.

Lexical screening cannot prove semantic novelty.

Agreement is not human ground truth.

Preserved exposed practice

Validated model-reference matches:

Llama 73/100; rules 18/100.

All 24 Llama validation errors remain counted.

Control success: 23.81% Llama, 8.40% rules,

3.92% item feedback.

NDCG@10: 0.02461 Llama vs 0.03128 feedback.

41,508 ranking rows; separate recount passed.

Exposed bank, not an unseen test.

Prespecified analysis

User is the analysis unit. Average three seeds within each request/class, then equal eligible classes per user.

5,000 paired user bootstrap samples, seed 2026.
Pratt one-sided Wilcoxon with joint BH over 11 tests.

Keep failures, infeasibility and exclusions visible.
Report control, HR@1 and NDCG@10 together.

Frozen prompts, validators, model digest, ranking inputs, seeds and metrics precede inference. No outcome tuning.

Fresh model-reference results

Validated command agreement: Llama 76/100;

rules 11/100; 23 Llama validation rejections.

Control success: Llama 20.145%, rules 3.854%,

item feedback 3.815%.

Conditional requests: zero successes.

NDCG@10: 0.02580 Llama vs 0.03184 feedback.

41,520 ranking rows; 2,595 score vectors.

Requests synthetic; references model-generated.

Contribution and limits

H1_AI and H2_AI: supported under the

frozen model-reference protocol.

Results do not establish conditional control.

Next-item ranking quality was lower with Llama.

Human experience and other models need new evidence.

Opaque categories and next-item proxy limit claims.

Interpretation of the results

Language translated richer multi-category requests.

Deterministic rules applied validated controls.

Pooled control gains came with lower next-item NDCG@10.

Conditional requests had no successful controls.

Synthetic wording and model references limit generalization.

Evidence boundaries

100 unique request texts were reused across 865 users.

References came from two model annotation contexts.

Public screening receipts include candidate texts.

Full structured cases and ranking rows remain local.

Human understanding and satisfaction were not measured.

Full Windows and Colab replay remains unverified.

Backup: metric definitions

TCP@10 Target-category items divided by ten for boost requests.

TCER@10 Target-category items divided by ten for mute requests.

HR@1 One if the first item equals the held-out target, otherwise zero.

NDCG@10 For one relevant item at rank $r \leq 10$, $1/\log_2(r+1)$. Zero if absent.

Both exposure metrics have the same arithmetic and opposite desired directions. Historical quality does not directly measure preference satisfaction after intervention.

Backup: backbone and control settings

Backbone	Compact SASRec-style implementation
Hidden size / context	32 / 50 items
Attention	Two blocks, one head
Training	20 epochs, Adam, learning rate 0.001
Batch / dropout	64 / 0.2
Training seeds	42, 43, 44
Experiment 1 control	Normalized score +0.25, or hard mute
Experiment 2 strengths	0.10, 0.25, 0.50
Language backend	Llama 3.1 8B, temperature 0

The backbone is independently implemented. The study does not claim an exact SASRec paper reproduction.

Backup: why non-inferiority is uninformative

Pooled Experiment 1 quantity	Value
A HR@1 baseline	0.2505%
C minus A, mean	$+0.193$ pp
Paired 95% interval	$[-0.058, +0.501]$ pp
Simultaneous lower bound	-0.07707 pp
Absolute proposed threshold	-10 pp
Relative 10% loss threshold	-0.02505 pp

The absolute reading numerically passes while allowing the entire small baseline to disappear. The relative reading fails. Only eight of 865 users have nonzero paired HR@1 differences.

Both operations pooled, distinct from the boost-only NDCG contrast. pp means percentage points.

Backup: intended-direction movement

$$D(S) = \frac{\sum_{r=1}^{10} \mathbf{1}[i_r \in S] / \log_2(r+1)}{\sum_{r=1}^{10} 1 / \log_2(r+1)}$$

ΔD is the channel value minus its uncontrolled value.

Class	Signed movement M
Compound	$\min(\Delta D(X), -\Delta D(Y))$
Boost / reduction	$\Delta D(X)$ / $-\Delta D(X)$
Floor	$-\Delta D(X)$, zero if the floor is violated
Conditional	$\Delta D(X \cap Y) - \max(0, \Delta D(X \setminus Y))$

Induced satisfaction requires a full list, the class band and $M > 10^{-12}$. This operational measure is not Shannon information.

Backup: reproducibility evidence

100 requests and 200 retained parser predictions.

3,460 assigned user/request pairs.

41,520 ranking rows; 2,595 score vectors.

Fixed prompt, model digest, seeds and input hashes.

Saved audit: 1,666,140 checks, no discrepancies.

Full Windows and Colab replay remain open.

Selected references

Mozilla (2022): Does this button work?

Ramos et al. (2024): natural-language profiles.

Sanner et al. (2023): language-based preferences.

Chen and Pu (2012): critiquing recommenders.

Kang and McAuley (2018): SASRec.

Gao et al. (2022): KuaiRec.

Full records and source links: paper/references.bib.